Triangles & Congruence

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Review: Equations of lines**

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

Final Answer: $x = 1$ and $x = 5$. Factoring $(x - 1)(x - 5) = 0$ yields the roots $x = 1$ and $x = 5$.

2 **[Foundational]** • **Target: Review: Equations of lines**

A data set recorded for review: equations of lines consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

3 **[Foundational]** • **Target: Review: Equations of lines**

For the model $f(x) = 6x + 3$ in review: equations of lines, evaluate $f(3)$.

Final Answer: 21. Substitute $x = 3$: $f(3) = 6(3) + 3 = 21$.

4 **[Foundational]** • **Target: Review: Equations of lines**

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

Final Answer: $x = 2$ and $x = 6$. Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

5 **[Foundational]** • **Target: Review: Equations of lines**

For the model $f(x) = 6x + 3$ in review: equations of lines, evaluate $f(4)$.

Final Answer: 27. Substitute $x = 4$: $f(4) = 6(4) + 3 = 27$.

6 **[Foundational]** • **Target: Review: Equations of lines**

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

Final Answer: $x = 3$ and $x = 7$. Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

7 **[Foundational]** • **Target: Review: Equations of lines**

For the model $f(x) = 6x + 3$ in review: equations of lines, evaluate $f(5)$.

Final Answer: 33. Substitute $x = 5$: $f(5) = 6(5) + 3 = 33$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Review: Equations of lines

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

Final Answer: $x = 4$ and $x = 8$. Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

9 [Intermediate] • Target: Equations of parallel and perpendicular lines

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 90°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (45^\circ + 45^\circ) = 90^\circ$.

10 [Intermediate] • Target: Equations of parallel and perpendicular lines

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 8 cm. Multiplying the original length by the scale factor gives $4 \times 2 = 8$ cm.

11 [Intermediate] • Target: Equations of parallel and perpendicular lines

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

Final Answer: 9. Substituting $x = 1$ into $y = 5x + 4$ gives $y = 5(1) + 4 = 9$.

12 [Intermediate] • Target: Equations of parallel and perpendicular lines

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 83°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (45^\circ + 52^\circ) = 83^\circ$.

13 [Intermediate] • Target: Equations of parallel and perpendicular lines

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 12 cm. Multiplying the original length by the scale factor gives $4 \times 3 = 12$ cm.

14 [Intermediate] • Target: Equations of parallel and perpendicular lines

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Final Answer: 14. Substituting $x = 2$ into $y = 5x + 4$ gives $y = 5(2) + 4 = 14$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 8.01 Area, Surface Area & Volume

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 76°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (45^\circ + 59^\circ) = 76^\circ$.

16 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 9.01 Transformations & Symmetry

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 16 cm. Multiplying the original length by the scale factor gives $4 \times 4 = 16$ cm.

17 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 1.01 Foundations & Reasoning

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.

Final Answer: 19. Substituting $x = 3$ into $y = 5x + 4$ gives $y = 5(3) + 4 = 19$.

18 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 10.01 Coordinate Geometry

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.

Final Answer: 69°. Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (45^\circ + 66^\circ) = 69^\circ$.

19 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 11.01 Probability

A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

Final Answer: 20 cm. Multiplying the original length by the scale factor gives $4 \times 5 = 20$ cm.

20 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 2.01 Lines & Angles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Final Answer: 24. Substituting $x = 4$ into $y = 5x + 4$ gives $y = 5(4) + 4 = 24$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Parallel lines and transversals

In ΔABC studying parallel lines and transversals, $m\angle A = 42^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 93° ✓

b. 103°

c. 85°

d. 138°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (42^\circ + 45^\circ) = 93^\circ$.

22 [Foundational] • Target: Parallel lines and transversals

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?

a. 15 cm

b. 18 cm ✓

c. 24 cm

d. 54 cm

Multiplying the original length by the scale factor gives $6 \times 3 = 18$ cm.

23 [Foundational] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$?

a. 12

b. 16

c. 14 ✓

d. 16

Substituting $x = 3$ into $y = 2x + 8$ gives $y = 2(3) + 8 = 14$.

24 [Foundational] • Target: Parallel lines and transversals

In ΔABC studying parallel lines and transversals, $m\angle A = 42^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 72° ✓

b. 82°

c. 64°

d. 138°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (42^\circ + 66^\circ) = 72^\circ$.

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25 [Foundational] • Target: Proving lines parallel

In ΔABC studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 92° ✓

b. 102°

c. 84°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 45^\circ) = 92^\circ$.

26 [Foundational] • Target: Proving lines parallel

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?

a. 18 cm

b. 21 cm ✓

c. 28 cm

d. 63 cm

Multiplying the original length by the scale factor gives $7 \times 3 = 21$ cm.

27 [Foundational] • Target: Proving lines parallel

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$?

a. 15

b. 20

c. 18 ✓

d. 21

Substituting $x = 3$ into $y = 3x + 9$ gives $y = 3(3) + 9 = 18$.

28 [Foundational] • Target: Proving lines parallel

In ΔABC studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?

a. 71° ✓

b. 81°

c. 63°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 66^\circ) = 71^\circ$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Parallel lines and transversals

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm ✓

c. 18 cm

d. 24 cm

Multiplying the original length by the scale factor gives $6 \times 2 = 12$ cm.

30 [Intermediate] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 12

c. 10 ✓

d. 12

Substituting $x = 1$ into $y = 2x + 8$ gives $y = 2(1) + 8 = 10$.

31 [Intermediate] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 10

b. 14

c. 12 ✓

d. 14

Substituting $x = 2$ into $y = 2x + 8$ gives $y = 2(2) + 8 = 12$.

32 [Intermediate] • Target: Parallel lines and transversals

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm ✓

c. 30 cm

d. 96 cm

Multiplying the original length by the scale factor gives $6 \times 4 = 24$ cm.

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33 [Intermediate] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 14

b. 18

c. 16 ✓

d. 18

Substituting $x = 4$ into $y = 2x + 8$ gives $y = 2(4) + 8 = 16$.

34 [Intermediate] • Target: Proving lines parallel

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 14

c. 12 ✓

d. 15

Substituting $x = 1$ into $y = 3x + 9$ gives $y = 3(1) + 9 = 12$.

35 [Intermediate] • Target: Proving lines parallel

In ΔABC studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85° ✓

b. 95°

c. 77°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 52^\circ) = 85^\circ$.

36 [Intermediate] • Target: Proving lines parallel

In ΔABC studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 78° ✓

b. 88°

c. 70°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 59^\circ) = 78^\circ$.

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37 [Intermediate] • Target: Proving lines parallel

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm ✓

c. 42 cm

d. 175 cm

Multiplying the original length by the scale factor gives $7 \times 5 = 35$ cm.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Review: Equations of lines • Cross-Lesson: 10.01 Coordinate Geometry

A data set recorded for review: equations of lines consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

39 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 11.01 Probability

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 11

c. 9 ✓

d. 14

Substituting $x = 1$ into $y = 5x + 4$ gives $y = 5(1) + 4 = 9$.

40 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 2.01 Lines & Angles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 9

b. 16

c. 14 ✓

d. 19

Substituting $x = 2$ into $y = 5x + 4$ gives $y = 5(2) + 4 = 14$.

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41 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 76° ✓

b. 86°

c. 68°

d. 135°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (45^\circ + 59^\circ) = 76^\circ$.

42 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm ✓

c. 24 cm

d. 100 cm

Multiplying the original length by the scale factor gives $4 \times 5 = 20$ cm.

43 [Challenging] • Target: Extension: Perpendiculars and distance • Cross-Lesson: 6.01 Polygons & Quadrilaterals

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

44 [Challenging] • Target: Extension: Perpendiculars and distance • Cross-Lesson: 7.01 Circles

For the model $f(x) = 3x + 5$ in extension: perpendiculars and distance, evaluate $f(3)$.

a. 14 ✓

b. 17

c. 9

d. 18

Substitute $x = 3$: $f(3) = 3(3) + 5 = 14$.

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45 [Challenging] • Target: Extension: Perpendiculars and distance • Cross-Lesson: 8.01 Area, Surface Area & Volume

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

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